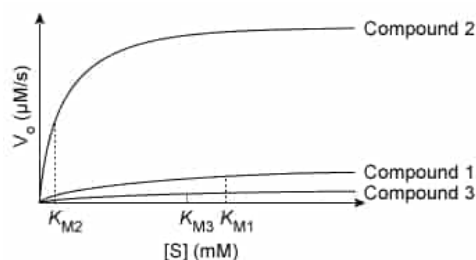


An enzyme was assessed for its ability to cleave three substrates with similar structures, Compounds 1–3, as shown.



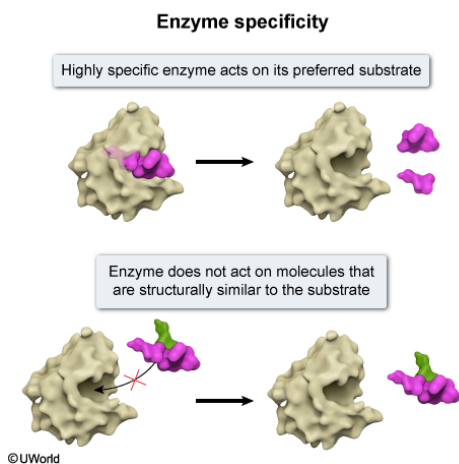
Based on the graph, which statement best describes the interaction of the enzyme with the compounds? The enzyme:

- ☐ A. is strongly inhibited by Compounds 1 and 3. [21%]
- ☐ B. is allosterically activated by Compound 2. [8%]
- ☒ C. has high specificity for Compound 2. [66%]
- ☐ D. has a high Hill coefficient for Compounds 1 and 3. [3%]

Correct

66% answered correctly

Explanation:



Some enzymes are **highly specific** for a particular substrate. These enzymes demonstrate **high reaction rates** with the **substrate of interest** but **not with any other molecules**, including those that are structurally similar to the substrate. For example, the α -amylase enzyme cleaves the glycosidic bonds in starch but not cellulose, even though the two molecules differ only in the **orientation** of their glycosidic bonds.

In the graph, the enzyme yields a high reaction velocity (V_o) with Compound 2 as the substrate but not with Compounds 1 or 3. Compound 2 also has a substantially lower K_M value than the other compounds.

Because K_M is an approximate measure of binding affinity, Compound 2 has a significantly higher affinity for the enzyme than the other compounds. In addition, the low V_{max} values of Compounds 1 and 3 show that the enzyme does not readily convert them to product. Therefore, the enzyme is **highly specific for Compound 2**.

(Choice A) Compounds 1 and 3 were tested as *substrates*, not *inhibitors*. To test their ability to inhibit the enzyme, each compound would have to be mixed with Compound 2, and the rate of Compound 2 cleavage would have to be assessed. In addition, although the structural similarities of Compounds 1 and 3 to Compound 2 makes them *candidates* for **competitive inhibition**, their relatively high K_M values suggest that

they have a *low affinity* for the active site. Therefore they would **not efficiently compete** with Compound 2 for binding, and would be *poor* inhibitors.

(Choice B) The question states that the three compounds tested are substrates, not **allosteric effectors**. The graph gives no information about allosteric interactions.

(Choice D) A high **Hill coefficient** indicates cooperativity and would result in a sigmoidal curve. Neither Compound 1 nor Compound 3 shows sigmoidal behavior.

Educational objective:

Some enzymes have high specificity for their substrates. These enzymes catalyze only reactions with a particular substrate and do not act on other molecules with similar, but not identical, structure.

Time spent:0 sec

Subject:
Biochemistry

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System:
Enzymes